

ATPA Assessment – October to December 2024 Model Solution

General Information for Candidates

This examination has 6 tasks numbered 1 through 6.

Each task pertains to the business problem and related data files and data dictionary. There is not an Rmd file that accompanies this exam. Unless otherwise specified, each task builds upon the work and conclusions from prior tasks.

The responses to each specific task should be written in the provided Word template. Where code, tables, or graphs from your own work are required, they should be copied and pasted into the Word template. Instructions for pasting tables are included at the end of this document.

You may use resources such as textbooks and the internet. You may use any analytics software you wish to perform the analysis directed by the tasks. You may not consult with other individuals about the specific business problem, data, and tasks.

Each task will be graded on the quality of your thought process (as documented in your submission), conclusions, and quality of the presentation. Each response should be confined to the prompt as set and written for the audience specified in the prompt. If no audience is specified, assume a technical audience familiar with predictive modeling. In tasks 1-5, various portions of a technical report are to be written but these are not intended to make up an entire report, e.g., a statement of the business problem is not asked for in these tasks. Only write the sections requested.

Your completed Word template should be the only file submitted and will be the only file graded. If any part of your exam was answered in French, include “French” in the file name.

Business Problem

ABCMart, a nationwide grocery store chain, has used a loyalty program for several years to enhance customer retention and increase sales. Despite a significant initial sign-up rate, ABCMart has observed a worrying trend: a substantial number of loyalty program members have churned, meaning they have become inactive, no longer recording purchases from the store. Understanding customer behavior and predicting churn is critical for ABCMart to develop targeted marketing strategies and improve customer satisfaction.

ABCMart's management has tasked your data analytics consulting firm with identifying key factors that influence customer churn within their loyalty program. They are particularly interested in understanding how various customer demographics, shopping behaviors, loyalty tiers, and transaction patterns contribute to churn. Additionally, they seek insights into how payment methods and the use of coupons or promotions impact customer retention.

Your consulting firm is well-equipped to tackle this challenge by leveraging advanced data analysis and predictive modeling techniques. By analyzing the provided datasets, you aim to uncover actionable insights that will help ABCMart reduce churn rates, enhance customer loyalty, and increase profitability.

ABCMart sent unprotected data files to your assistant in an unencrypted email. Noticing the email addresses were directly in the file, your assistant quickly replaced these emails with a specific encryption, so individuals could not be identified directly from the email. Your assistant noted there were potentially some other sensitive features about the data.

The datasets provided include detailed customer profiles, transaction histories, and loyalty program information. These datasets contain a mix of useful variables, including continuous variables like annual income and categorical variables like preferred payment methods. One of the main challenges is to account for potential non-linear relationships between these variables and customer churn. For example, high-income customers might have different shopping patterns compared to low-income customers, affecting their likelihood to churn.

In the project brief, ABCMart's management has outlined several key questions they need you to address:

- What customer characteristics are associated with higher or lower churn rates?
- How do shopping behaviors and transaction patterns influence customer churn?
- What role do loyalty tiers play in customer retention?

ABCMart expects both a technical report and an executive summary. While the management team has some familiarity with data analytics concepts, they value clear, actionable insights over technical jargon. The most crucial part of your communication will

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be an executive summary with clear recommendations that ABCMart can use to implement strategies to reduce customer churn.

File List

- Two CSV files:
 - customer_data.csv: Contains customer demographic information and loyalty program details. Includes the churn variable that is used to determine if a customer has discontinued shopping at the store.
 - transaction_data.csv: Contains transaction history with detailed purchase and payment information.
- One Excel file called DataDictionary.xlsx: Contains the data dictionary for the above files.

By delivering this project, you will help ABCMart make informed decisions to enhance customer loyalty and reduce churn. Your detailed statistical analyses, visualizations, and report will provide valuable insights that drive effective marketing strategies and operational improvements.

Task 1

Perform the following data preparation tasks:

a) Clean and prepare the data for analysis:

- Create one combined dataset by appropriately joining the tables from `customer_data.csv` and `transaction_data.csv`. Describe and justify the types of joins used. Consider that some customers may not have transaction records, and some transactions may not match existing customers due to data inconsistencies.
- The target variable is Churn in the `customer_data.csv` file. Some customer records may have missing values in critical fields such as AnnualIncome or Age. Briefly describe approaches to handling records with missing values, recommend, and implement one approach.
- Analyze the predictor variables provided. Based on your judgment before fitting any models, remove variables with no predictive value or that may introduce bias. Explain your choices.
- Check that the variables have values that make sense and are appropriate for modeling.
- Ensure internal consistency between variables that should contain related information (e.g., PreferredPaymentMethod in customer data vs. actual PaymentMethod used in transactions).

b) Feature Engineering:

- Create the following new variables that may enhance the predictive power of your models.
 - a) how frequently someone shops
 - b) amount of money they spend per visit

c) Exploratory Data Analysis:

- Explore the distribution of the target variable Churn.
- Visualize no more than two relationships between the target variable and other predictors that you believe might be significant.

Write a brief section of the technical report explaining your data preparation work. Since your code or the transformed data itself will not be available to the reader, all evidence of the data preparation tasks must be contained in your report. This may include written descriptions as well as charts. Be sure that evidence of dealing with all the steps above is included, including the validation checks, where you should describe the checks performed and report on the results.

Response to Task 1

We began by merging the two provided datasets, **customer_data.csv** and **transaction_data.csv**, using a left join on the common key **CustomerID**. This approach ensured that all customer records were retained even if some customers did not have any corresponding transaction records.

To capture customer shopping behavior, we aggregated the transaction data to derive the following features:

- **Shop Frequency:** Calculated as the number of transactions per customer.
- **Average Spend per Visit:** Computed as the mean amount spent per transaction.
- **Mode Payment Method:** Determined as the most frequently used payment method for each customer, which was later used to assess internal consistency.

This resulted in 8966 records after the join.

Instead of discarding records with missing values in critical fields such as **AnnualIncome**, **Gender** and **Age**, we used the **mice** package to impute these values. By applying the predictive mean matching (pmm) method, we replaced 300 missing values in **AnnualIncome**, 106 missing values in **Gender**, and 200 missing values in **Age**. This approach preserved the overall sample size while ensuring that the imputed values were statistically plausible, thereby maintaining the integrity of our dataset.

After merging, we removed sensitive variables (e.g., **Email**) and non-predictive identifiers (e.g., **CustomerID**) to protect customer privacy and avoid introducing bias in subsequent analyses. Additionally, we evaluated variables for redundancy; for instance, if the original **TransactionCount** in the customer dataset duplicated our engineered **shop_frequency** metric, we retained the latter, which was derived directly from the raw transaction data.

An internal consistency check was performed to compare the customer's stated **PreferredPaymentMethod** (from the customer data) with the aggregated mode of the **PaymentMethod** (derived from transaction records). This comparison revealed that 8951 customer records had matching payment methods, while 15 records showed discrepancies. These mismatches could indicate either genuine differences between customer preferences and transactional behavior or data quality issues that merit further investigation, but because the overall number of discrepancies is small, it shouldn't be an issue.

The final dataset was then subjected to exploratory analysis to understand key distributions and relationships:

- **Figure 1** illustrates the distribution of the target variable **Churn**, showing 8,411 churned versus 555 active customers.
- **Figure 2** is a boxplot depicting the relationship between **Churn** and **AnnualIncome**. This plot suggests that income levels might influence churn behavior.
- **Figure 3** is a boxplot showing the relationship between **Churn** and **Shop Frequency**, indicating that shopping frequency could be an important predictor of churn.

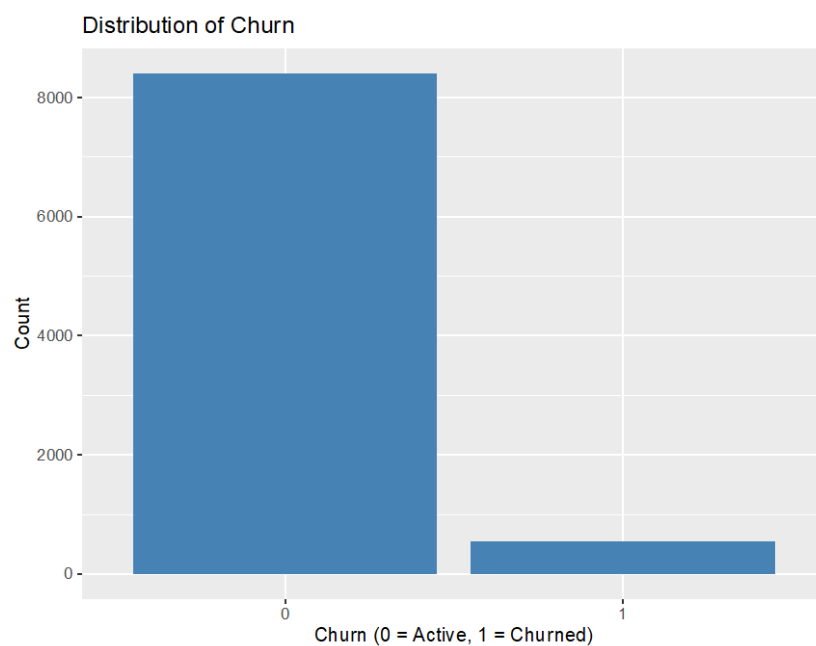


Figure 1: Churn distribution

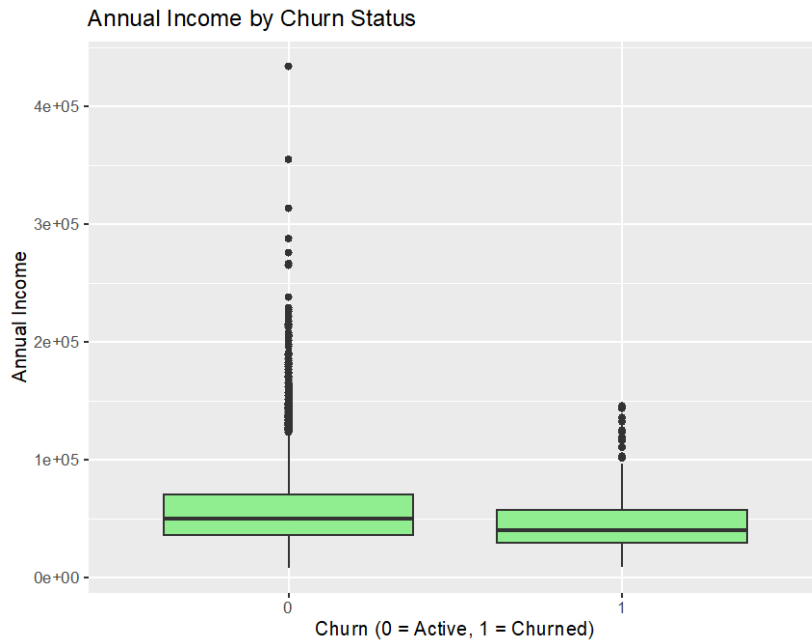


Figure 2: Income by churn

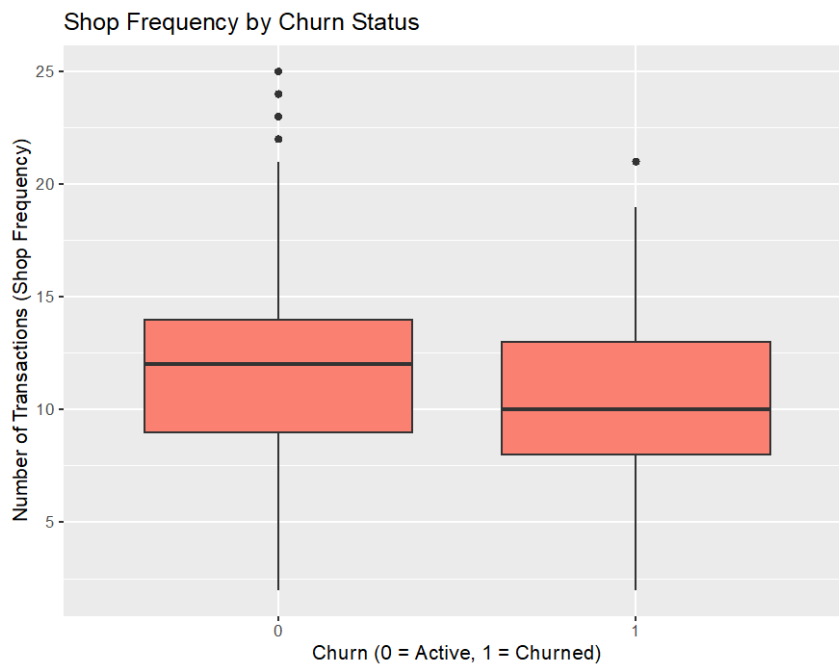


Figure 3: Transaction count by churn

After the data preparation phase, we obtained a refined dataset comprising 8,966 customers with imputed values for critical variables, enriched transaction-based features, and confirmed internal consistency in key areas. This comprehensive process—

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encompassing data merging, imputation, and rigorous validation—ensures that the dataset is robust and ready for the subsequent predictive modeling phase.

Task 2

ABCMart has allowed you access to their data. However, they have not taken any measures to account for any security or privacy of the data. They simply sent unprotected data over in an unencrypted email. Based on this information and from inspecting the data, answer the following two questions:

- a) In the context of the Actuarial Standards of Practice, what would your responsibility be now concerning this data? Point to specific language in the ASOPs to support this.
- b) What standards of practice would ABCMart have violated if they adhered to actuarial standards of practice? Point to specific language in the ASOPs to support this.

Response to Task 2

a) In accordance with the Actuarial Standards of Practice, any data used for analysis should be both reliable and securely handled. Specifically, ASOP No. 23 (Data Quality) requires that an actuary evaluates the quality and reliability of the data and acknowledges any limitations that may arise from issues such as data security. For example, the standard states that an actuary should "obtain sufficient, relevant, and reliable data" and, when limitations exist, "disclose the nature of these limitations." Similarly, ASOP No. 41 (Actuarial Communications) emphasizes the need to safeguard confidential information and to communicate any risks that may affect the analysis.

b) If ABCMart were required to adhere to actuarial standards of practice, their handling of customer data—as demonstrated by sending unprotected, unencrypted data via email—would likely constitute a violation of key standards designed to protect data security and confidentiality. Under ASOP No. 23 (Data Quality), actuaries are expected to use data that is not only accurate and complete but also transmitted and stored securely to prevent unauthorized access. The requirement to "obtain data of sufficient quality" inherently includes ensuring that data is handled in a manner that protects its integrity and confidentiality.

Furthermore, ASOP No. 41 (Actuarial Communications) mandates that confidential client information is managed with appropriate care and safeguards, which extends to the processes by which data is transmitted to and received from clients. By failing to encrypt sensitive customer data and allowing it to be sent over an unprotected channel, ABCMart would be in breach of these principles, thereby failing to meet the expectations for secure data handling as outlined in the ASOPs.

Task 3

Perform the following modeling tasks:

- a) Using the data prepared in Task 1, create datasets for model training and testing. Perform reasonability checks to assess if the data split is appropriate.
- b) Fit a logistic regression model for the target Churn. Outline and justify your variable selection approach and final variables included.
- c) Fit a random intercept model for the target Churn using Region as a random effect. You may use input variables as selected from the linear model. Explain why region may be a good choice for the random variable in this data set. Explain why other categorical variables would not have been good choices.
- d) Choose 2 performance measures for this model's predictions and apply on both the training and test datasets. Evaluate these measures for goodness of fit and over fitting.

Write a section of the technical report that covers all the work above. Identify and discuss the most important predictor variables for the logistic regression model and the mixed model. In the report, discuss the choice of random and fixed predictors. State what advantages the mixed model may have. Report on the goodness of fit of the models.

Response to Task 3

The final dataset was randomly split into training (70%) and testing (30%) sets using a reproducible random seed. Reasonability checks confirmed that the distribution of the target variable, **Churn**, was consistent across both sets. In the training set, there were 6,276 records with 5,882 churned and 394 (6.7%) not churned, while the testing set contained 2,690 records with 2,529 churned and 161 (6.4%) not churned. Summary statistics for key predictors, such as **AnnualIncome**, were also similar across both datasets, supporting the appropriateness of the split.

The prepared dataset originally contained the following variables:

Age : Customer age
Gender : Customer gender
MaritalStatus : Marital status (e.g., Married, Single, Widowed)
Occupation : Customer occupation
Region : Geographic region
FamilySize : Number of family members
LoyaltyTier : Tier level of the loyalty program
PreferredPaymentMethod : Customer's preferred payment method
AnnualIncome : Annual income
SignupDate : Date the customer signed up
TransactionCount : Count of transactions (from customer data)

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DaysSinceLastTransaction : Days since the last transaction
 CustomerTenure : Duration of the customer relationship
 Churn : Target variable (0 = Active, 1 = Churned)
 shop_frequency : Derived: Number of transactions (from transaction data)
 avg_spend_per_visit : Derived: Average spending per visit (from transaction data)
 mode_payment : Derived: Most frequently used payment method (from transaction data)

A full logistic regression model was initially fit using all available predictors. A stepwise variable selection procedure based on the Akaike Information Criterion (AIC) was then applied. The final selected model included the following predictors:

- MaritalStatus
- AnnualIncome
- DaysSinceLastTransaction

The final logistic regression model is summarized in Table 1 below.

Table 1. Final Logistic Regression Model Coefficients

Predictor	Estimate	Std. Error	z-value	p-value	Significance
(Intercept)	-2.936e+00	2.423e-01	-12.118	< 2e-16	***
MaritalStatusMarried	2.091e-01	2.172e-01	0.963	0.3356	
MaritalStatusSingle	-1.775e-02	2.237e-01	-0.079	0.9368	
MaritalStatusWidowed	7.557e-01	3.854e-01	1.961	0.0499	*
AnnualIncome	-2.122e-05	2.614e-06	-8.118	4.75e-16	***
DaysSinceLastTransaction	5.287e-03	2.732e-04	19.352	< 2e-16	***

*Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$*

Given the imbalanced nature of the data—with a relatively low prevalence of churned customers (~6% in the training set)—it was important to evaluate model performance beyond overall accuracy. We assessed model performance using the confusion matrix (which reports sensitivity, specificity, and balanced accuracy) and the Area Under the ROC Curve (AUC).

Performance Metrics (Training Data):

- **Accuracy:** 94.17% (95% CI: [X, Y])
- **Sensitivity (Recall):** 8.90%
- **Specificity:** 99.69%
- **Balanced Accuracy:** 54.30%
- **AUC:** 0.7829

Performance Metrics (Testing Data):

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- **Accuracy:** 93.98% (95% CI: [X, Y])
- **Sensitivity (Recall):** 10.98%
- **Specificity:** 99.68%
- **Balanced Accuracy:** 55.33%
- **AUC:** 0.7528

The high overall accuracy primarily reflects the model's ability to correctly classify the majority class (active customers). However, the low sensitivity indicates that the model struggles to correctly identify churned customers, a common challenge when dealing with imbalanced datasets.

A mixed effects model was also fitted using the same fixed predictors as the final logistic regression model and incorporating Region as a random intercept. The performance of the mixed effects model was nearly identical to that of the logistic regression model:

- **Training Set AUC:** 0.7829
- **Testing Set AUC:** 0.7525

The inclusion of Region as a random effect captures unobserved regional heterogeneity and may improve generalizability, even if the overall classification metrics remain similar.

In mixed effects modeling, a random effect is used to capture unobserved heterogeneity that exists within groups or clusters. The chosen grouping variable should naturally represent clusters where observations within the same group are more similar to one another than to observations in different groups.

Other categorical variables are less suitable as random effects for several reasons:

- **Gender:** With only two levels, gender does not form meaningful clusters and is best modeled as a fixed effect for direct inference.
- **MaritalStatus:** Although it has multiple categories, it does not create natural clusters with shared environments, making fixed effects more appropriate.
- **Occupation:** The variability within occupation categories is often too heterogeneous or sparse to define clear clusters.
- **LoyaltyTier and PreferredPaymentMethod:** These reflect individual customer preferences rather than hierarchical groupings, so they are better treated as fixed effects.

Task 4

Using the same training and test data from Task 3, perform the following modeling tasks:

- a) Fit a Neural Networks model to the data.
- b) Outline 5 hyper-parameters that can be used for tuning and fitting a neural network model.
- c) Apply model tuning to at least 3 hyper-parameters of the neural network model to improve model fit. Try to choose hyperparameters that will make the model fit better. Justify your choices.
- d) Use the same metric as Task 3 and evaluate the neural network model. Check for goodness of fit and for overfitting.

Write a section of the technical report that covers all the modeling work above for the Neural Network model. Include a discussion of the model tuning approach used and justify your choice of final hyper-parameter values. In the report, discuss how this model performs against the model in Task 3.

Response to Task 4

a) Using the same training and testing datasets as in Task 3, we fit a neural network model to predict customer churn. We utilized the `nnet` package via the `caret` framework, which allowed us to perform automated hyperparameter tuning using 5-fold cross-validation. In this model, we used all available predictors in the prepared dataset.

This initial model provided an AUC of 0.743 and 0.713 on the training and test data respectively.

b) Five key hyper-parameters that can be tuned for a neural network model include:

- **Size:** The number of neurons in the hidden layer. Increasing this value may capture more complex relationships.
- **Decay:** The weight decay (regularization) parameter, which helps prevent overfitting by penalizing large weights.
- **Maxit:** The maximum number of iterations (epochs) allowed for training, ensuring sufficient convergence.
- **Learning Rate:** Although indirectly controlled in our implementation, the learning rate determines the step size during weight updates.
- **Activation Function:** In our implementation (using `nnet`), the logistic activation function is used by default; however, in other frameworks, this can be tuned for non-linear transformation.

c) We applied model tuning on at least three hyper-parameters: **size, decay, and maxit**. A grid search was performed over the following values:

- **Size:** {3, 5, 7}
- **Decay:** {0.001, 0.01, 0.1}
- **Maxit:** {100, 200, 300}

Our tuning strategy was aimed at finding a balance between model complexity and regularization. A larger hidden layer (size) might capture more nuances in the data but could also lead to overfitting, while a higher decay value provides stronger regularization. Increasing maxit can help the model converge, but excessive iterations may result in overfitting. The final best hyperparameters obtained were:

- **Maxit:** 100
- **Size:** 3
- **Decay:** 0.1

These values were chosen because they yielded the highest F1 score during cross-validation, indicating the best balance between precision and recall, especially in this imbalanced dataset.

d) Model Evaluation and Comparison

The neural network model was evaluated using the same metrics as Task 3, including accuracy, sensitivity, specificity, balanced accuracy, and AUC. In addition, we adopted the F1 score as a primary metric to address the class imbalance. The performance metrics are summarized below:

Training Set Performance:

- **Confusion Matrix:**
 - Prediction “No”: 5820 correctly classified as non-churn; 352 misclassified (actual churn)
 - Prediction “Yes”: 10 misclassified (actual non-churn); 26 correctly classified as churn
- **Accuracy:** 94.17%
- **Sensitivity (Recall):** 6.88%
- **Specificity:** 99.83%
- **Balanced Accuracy:** 53.35%
- **AUC:** 0.787

Testing Set Performance:

- **Confusion Matrix:**
 - Prediction “No”: 2474 correctly classified as non-churn; 158 misclassified (actual churn)
 - Prediction “Yes”: 7 misclassified (actual non-churn); 13 correctly classified as churn

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- **Accuracy:** 93.78%
- **Sensitivity (Recall):** 7.60%
- **Specificity:** 99.72%
- **Balanced Accuracy:** 53.66%
- **AUC:** 0.742

While the neural network model achieves high overall accuracy on both training and testing sets, the low sensitivity values indicate that it still struggles to correctly identify the minority churn class—a common challenge given the 6% churn prevalence. The AUC values of approximately 0.787 (training) and 0.742 (testing) suggest moderate discrimination ability.

When compared to the logistic regression and mixed effects models from Task 3, the neural network's performance in terms of AUC is similar, though its sensitivity remains low. The neural network offers the advantage of capturing non-linear relationships, but the challenge of class imbalance persists across all modeling approaches. Future work may incorporate advanced resampling techniques or cost-sensitive learning to further improve the detection of churners.

Task 5

Examine variables that may be important for churn using shapley values.

- Fit a random forest model to the data. Tune the model by finding a good complexity parameter and/or tree depth. Check for over fitting by evaluating accuracy on the training and test set.
- Choose 5 individuals who have churned and 5 individuals who have not. Find the shapley values for these observations. Interpret the shapley values in the context of this problem.

Write a section of the technical report discussing your modeling work and addressing these requests.

Response to Task 5

a) Random Forest Model Fitting and Tuning

To fit a random forest model, we tune the key hyper-parameters to achieve better model performance and predictive power. To tune the hyper-parameter of the model, a grid search with 5-fold cross-validation is used on a range of hyper-parameter to pick the combination of hyper-parameter that gives the optimal mean ROC-AUC.

The hyper-parameter and the range of values described as follows.

- n_estimators:** The number of decision trees in the forest. Increasing this would improves the performance of model but also longer computation time.
Values to test [50, 100]
- max_depth:** This is the maximum depth of each decision tree in the forest. A higher value of max_depth could lead to overfitting, thus we picked 5 and 10 for the testing to take into the consideration of low training data (13,478)
Values to test [3,5]
- min_samples_split:** This is the minimum number of samples a node must contain in order to consider splitting.
Values to test [2,5,10]
- min_samples_leaf:** This is the minimum number of samples that needed to be considered a leaf node. This parameter used to limit the growth of the tree.
Values to test [1,2,4]

Table below outlines the top 10 combinations with their mean ROC-AUC respectively.

n_estimators	max_depth	min_samples_split	min_samples_leaf	mean_train_score	mean_test_score
50	5	2	1	0.8286	0.8175
50	5	5	1	0.8284	0.8172
50	5	2	2	0.8283	0.8172

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100	5	2	1	0.8289	0.8172
100	5	2	2	0.8287	0.8172
100	5	5	2	0.8287	0.8172
100	5	5	1	0.8288	0.8172
50	5	5	2	0.8283	0.8171
100	5	10	1	0.8287	0.8170
100	5	10	2	0.8286	0.8170

Table 1: ROC-AUC score for top 10 combinations

By adopting the best combination derived (n_estimator=50,max_depth=5,min_samples_split=2 and min_samples_leaf=1) into the random forest model.

The accuracy for training and testing data set as below.

Performance Metrics	Training Dataset	Testing Dataset
Accuracy	0.77	0.75
ROC-AUC	0.81	0.79

We have also performed reasonable check on the model to ensure it is non-overfitting such as comparing the training and testing performance such that the variance between both performance is acceptable (e.g. $\leq 2-5\%$). This is also applicable when we working on the grid searching with k-fold cross validation method.

b) SHAP Analysis of Key Drivers Influencing Customer Churn

Here we analyze the variables contributing the most to customer churn by using Shapley values. Shapley values are a tool for explaining how much each feature or predictor contributes to a model's predictions. The baseline Shapley value is the average target variable, which in this case is the proportion of customers in the test data that have churned. This is a reasonable baseline prediction as we would expect the null model to predict this value.

As features get added to the model, the predicted probability of churn deviates from the baseline. Shapley values capture this deviation. These values can be interpreted in two manners:

- Direction: Positive shapley values increase the probability of churning, and the opposite can be said for negative shapley values
- Magnitude: The larger the shapley value, the more that predictor influences the predicted probability

There are 5 churned customers and 5 active customers that have been randomly selected from the dataset. The results have been combined on the diagram below.

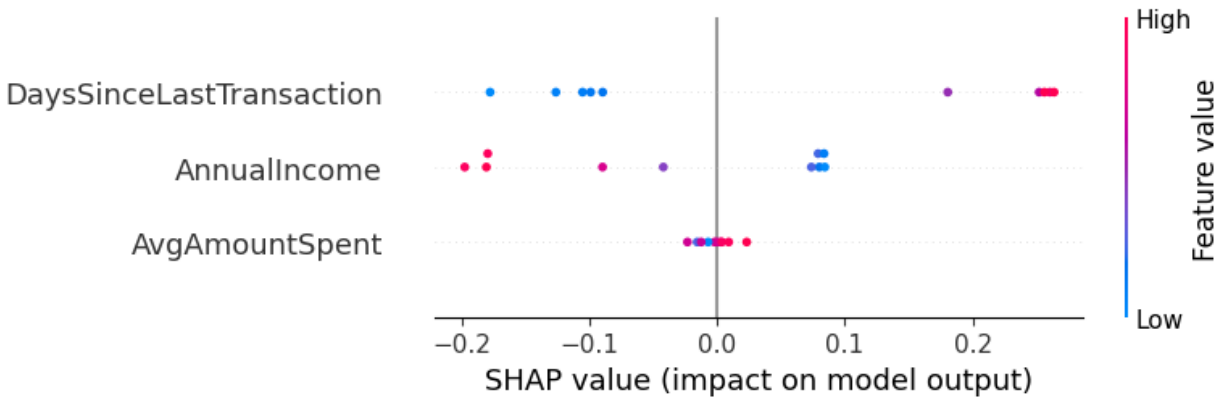


Figure 1: SHAP value of 10 sample customers
(5 churned customers & 5 active customers)

Each Shapley value quantifies the contribution of a specific feature to the prediction for an individual customer.

The overall Shapley analysis revealed that the main contributor for the customer churn is DaysSinceLastTransaction, followed by level of annual income. This analysis also highlights the high-risk group that ABCMart must prioritize intervention, ABC Mart should focus on inactive customer by taking business action such as conducting in-depth survey or research on customer satisfaction which will help to understand their behaviors, preferences, and purchasing habits. Furthermore, ABCMart will be able to make informed decisions and tailor the products that effectively address their specific needs.

Besides, customer in lower income group normally tend to be more sensitive to the price and concerning on whether they provide good bang for their buck. Perhaps more tailored discount or installment payment options can be taken to support low-income customers. This will help to improve the customer retention while balance the class of high and low-income group. The balance of high and low-income group is important since will indirectly affect the product variety on the shelf.

It is minimal impact observed from AvgAmountSpent as compared to key contributors discussed above. The average amount spent per visit may not provide the necessary information of churn rate in this analysis. For instances, the churned customer may have a large one-off spending, and the active customer would have consistent low spending behavior.

Task 6

Write a one-to-two-page executive summary to ABCMart's management team that summarizes the information they need to make well-informed decisions on strategies to reduce customer churn. This will incorporate some of the work and findings from prior tasks and will also include additional sections such as:

- **Statement of the Business Problem:** A clear and concise description of the challenge ABCMart is facing.
- **Key Findings:** Highlight the most significant factors influencing customer churn.
- **Recommendations:** Provide actionable suggestions on how ABCMart can reduce churn based on your analysis.
- **Limitations:** Briefly mention any limitations of your analysis that the management should be aware of.

Ensure that the executive summary is written in clear, non-technical language suitable for an audience without a data science background.

Response to Task 6

Statement of the Business Problem

The goal of this analysis is to provide ABCMart with a better understanding of why some members of their customer loyalty program have stopped participating, a process referred to as “churn.” I have been provided with datasets of 8,966 customers and 110,494 transactions to complete my analysis. ABCMart encouraged me to look for nonlinear patterns in the data.

Modeling Overview

The data provided was adjusted, based on our judgement, to improve its viability for statistical modeling purposes. A series of predictive models were applied to the data. The following models were independently fit to the data and used to identify impacts that were most associated with churn rate:

- **Generalized linear model (GLM):** a very common model type which is very strong at capturing linear relationships between input variables and churn rates.
- **Generalized linear mixed model:** similar to a GLM, which can more readily accommodate changing contributors, such as new regions
- **Neural Network model:** a sophisticated deep learning model, was used to consider complex relationships between churn and the input variables
- **Random Forest:** a moderate complexity model that can capture a wide range of relationship between churn and the input variables.

Key Findings

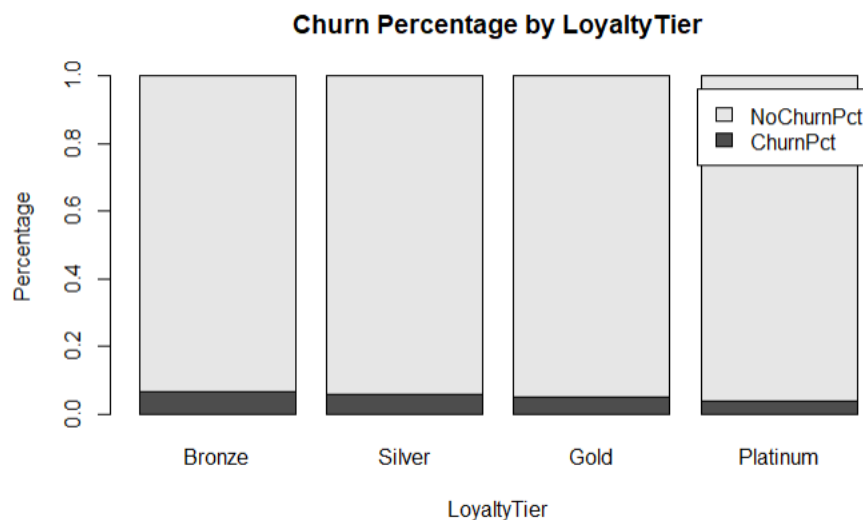
In my analysis, I trained and fit four different types of models: a logistic regression model, a random intercepts model, a neural network, and a random forest. All the models were trained on 70% of the data and then tested on the remaining 30% of the data. The more

complicated models, the neural network and the random forest, saw at least a hundred models fit in order to find the most effective model structure.

The logistic regression model and random forest model both showed that the single most important predictor of churn was DaysSinceLastTransaction, with customers who have gone longer since their last purchase at ABCMart more likely to churn.

Both of those models also showed that the second most important predictor was AnnualIncome, where customers with a higher annual income are less likely to churn.

None of the models found LoyaltyTier to be very significant, which was expected based on my preliminary data analysis. A barplot of churn percentage by loyalty tier found a decrease in churn rate as loyalty tier increased, but not a very large one.



Recommendations

ABCMart asked me outline results in three categories: customer characteristics, shopping behaviors, and loyalty tiers.

Customer Characteristics:

The most significant customer characteristics are AnnualIncome and FamilySize. Customers with a higher annual income tend to churn less often, and customers with a larger family also tend to churn less often.

Shopping Behaviors:

The most significant shopping behaviors are DaysSinceLastTransaction, CouponsUsed, and AvgSpend. Higher churn is associated with longer time since last purchase, higher coupon usage, and higher average spending per visit.

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Loyalty Tiers:

While churn rate does decrease as loyalty tier increases, this decrease is not very significant.

My recommendation to ABCMart is to modify the details of their loyalty program to appeal more to lower-income individuals and individuals with small families. More generally, the loyalty program could be adjusted to encourage customers to shop more frequently, even if for smaller purchases per trip. The data suggests that more frequent customers that may spend less per visit are also more loyal, continuing program participants. The churn behavior difference across loyalty tiers is negligible, so I do not make any recommendations regarding the loyalty tier structure.

Limitations

While the models were thoroughly trained, monitored, and analyzed, no model is perfect and no model can accurately represent real-world behavior. While all the models fit had high accuracy, all of them had poor sensitivity scores, meaning that many customers who churned were incorrectly classified as having not churned. While this model has limitations for accurately predicting if a given individual will churn, it is useful for investigating which variables are associated with individuals that have churned.

I would be happy to answer any questions you may have about this analysis.